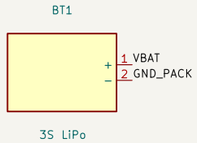
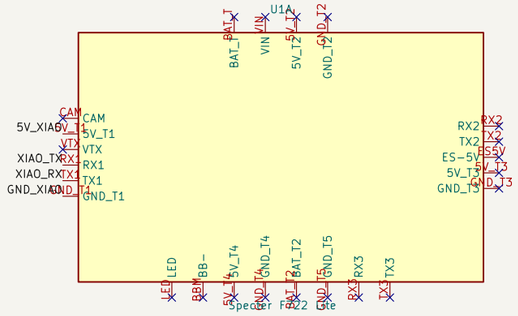
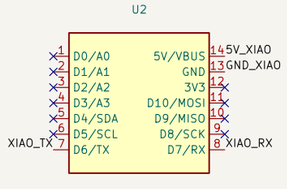
ARTICLE 2 HARNESS — net labels, not drawn wires. Netlist of record: harness-spec.md §6.

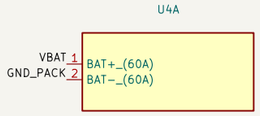
PROPOSED / NOT BUILT. Items marked (!) in harness-spec.md §8 are unverified.

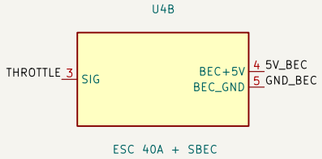


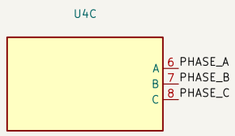


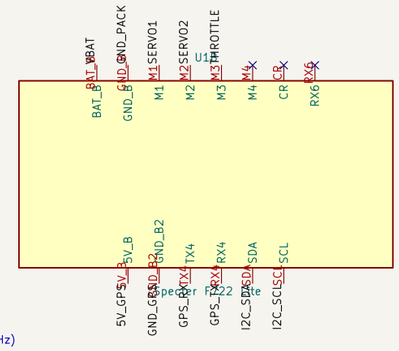


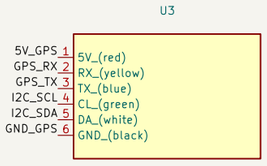
(!) XIAO 5V pin IS USB VBUS. FC rail plus a powered USB cable backfeeds the host port.







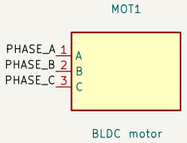


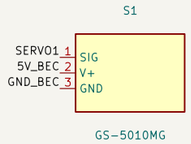


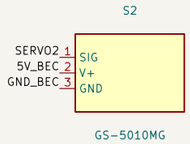
GPS FORWARD connector, left to right:

red 5V | yellow RX | blue TX | green CL | white DA | black GND

(RX before TX — confirmed on the module.)







U1 jumpers: ELRS short-connect pad = BRIDGED (internal RX on UART2).

UART1 = xiao MSP UART2 = ELRS CRSF UART3 = spare UART4 = GPS

UART5 has no pads. UART6 is RX-only. M1-M4 are the only output pads.

U1 unit 1 = TOP face pads. U1 unit 2 = BOTTOM face pads (ESC row + GPS row).

GROUND TOPOLOGY — star point is the ESC pack-negative pad. Four distinct WIRES, bonded only INSIDE a module:

GND_PACK BT1 − -> U4 BAT-(60A). U1 GND_B taps the same pads. Carries motor return (0-40 A).

GND_BEC U4 BEC_GND -> S1/S2 GND. Carries servo return (~2 A peak). Bonded to GND_PACK INSIDE U4.

GND_XIAO U1 GND_T1 -> U2 GND. GND_GPS U1 GND_B2 -> U3 GND. Both bonded to GND_PACK INSIDE U1.

(!) Do NOT take servo V+/GND from the FC — that routes servo return through the gyro's own voltage reference.

(!) Do NOT land the ESC lead's BEC_GND on an FC pad — same problem, plus a loop. Split the lead: only SIG goes to U1.

PROPOSED — not built, not verified on hardware
Grounds are 4 distinct wires, bonded only inside modules
Interconnect only — net labels, no drawn wires
autoc

Sheet: /
File: article2-harness.kicad_sch

Title: Article 2 — FC / xiao / GPS / ESC harness

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 1/1